NOT MEASUREMENT SENSITIVE

MIL-PRF-32405A (MR)
27 August 2021
SUPERSEDING
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06 January 2013

PERFORMANCE SPECIFICATION

CLEANER, HAND WIPE, METALLIC SUBSTRATES, LOW OR EXEMPT VOC AND HAP FREE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand wipe cleaners that do not contain hazardous air pollutants (HAP) (Clean Air Act 112(b)) and contain a low content of volatile organic compounds (VOC) or an exempt VOC (40 CFR 51.100) for use on DoD assets.

1.2 Classification. The hand wipe cleaner type and class will be as specified below.

1.2.1 Classes. The cleaner will be furnished in the following classes as specified (see 6.2).

Class 1 - Aqueous (contains water).

Class 2 - Non-aqueous, non-halogenated solvents.

Class 3 - Non-aqueous, halogenated solvents.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 6850

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1.2.2 Types. The types of hand wipe cleaners are as follows:

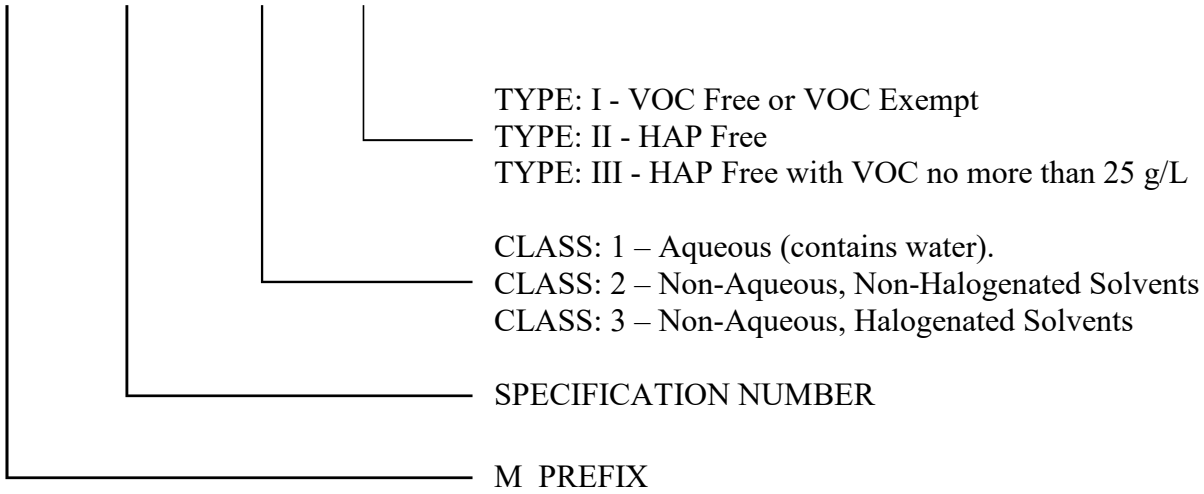
Type I - VOC and HAP free or VOC exempt and HAP free

Type II - HAP free

Type III - HAP free with VOC no more than 25 g/L

1.3 Part or Identifying Number (PIN). PINs to be used for cleaners acquired by this specification are created as follows:

M 32405 1 II



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

A-A-58054 - Abrasive Mats, Non-Woven, Non-Metallic

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FEDERAL STANDARDS

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-81706	-	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-PRF-23377	-	Coating, Epoxy, High-Solids
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-PRF-81322	-	Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code G-395
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne

(Copies of these documents are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil.)

2.3 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 51.100	-	Air Emissions Reporting Requirements
40 CFR 63	-	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 261	-	Identification and Listing of Hazardous Waste
40 CFR 401.15	-	Protection of Environment (Toxic pollutants)

(Copies of this document are available online at https://www.ecfr.gov/.)

2.4 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated (see 6.2), the issue in effect on date of invitation for bids or request for proposal should apply.

ASTM INTERNATIONAL

ASTM D740	-	Standard Specification for Methyl Ethyl Ketone

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ASTM D1193	-	Standard Specification for Reagent Water
ASTM D2109	-	Standard Test Methods for Nonvolatile Matter in Halogenated Organic Solvents and Their Admixtures
ASTM D3167	-	Standard Test Method for Floating Roller Peel Resistance of Adhesives
ASTM D3359	-	Standard Test Methods for Measuring Adhesion by Tape Test
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM F22	-	Hydrophobic Surface Films by the Water-Break Test
ASTM F485	-	Standard Test Method for Effects of Cleaners on Unpainted Aircraft Surfaces
ASTM F502	-	Standard Test Method for Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces

(Copies of these documents are available from https://www.astm.org/.)

SAE INTERNATIONAL

AMS QQ-A-250	-	Aluminum Alloys, Plate and Sheet (2024 & 7075)
AMS 6359	-	Steel, Sheet, Strip, and Plate 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)
AMS-H-6875	-	Heat Treatment of Steel Raw Materials

(Copies of these documents are available from https://www.sae.org.)

2.5 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The cleaning compound furnished under this specification shall be a product which has been tested by an ISO 9000/IEC 17025 or NADCAP certified 3rd party testing facility that is approved by the qualifying activity and has passed the qualification tests specified herein and has been listed or approved for listing on the applicable Qualified Products Database (QPD) by the qualifying activity (see 3.1.1, 3.1.2, 4.2, 4.3, 4.4 and 6.3) before contract award. Any change in the chemical formulation, the material, the process, or the procedure in manufacturing the cleaning compound shall necessitate its being qualified again. Any change requires a complete requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the products receiving qualification. Any cleaning compound that does not conform to all the qualification tests specified herein shall be removed from the QPD.

3.1.1 Retention of qualification. To retain qualification approval of products listed on the Qualified Products Database (QPD), the manufacturer shall be required to verify by certification

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to the qualifying activity that its product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity (see 6.3), the time of periodic verification by recertification shall be in two-year intervals from the date of original qualification (see 4.3). The recertification action is initiated by the qualifying activity.

3.1.2 Qualifying activity. The activity responsible for the collection and storage of cleaning compounds certifications that are required to be furnished under this specification for qualification and conformance testing is the Coatings and Corrosion Team, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

3.2 Material. The materials used in the cleaning compound that are supplied under this specification shall be as specified herein. The materials not specified shall be selected by the contractor and shall be subjected to all the provisions of this specification.

3.2.1 Composition. The composition of the cleaning compound shall be the same as that of the sample furnished for qualification testing. Any change in the formulation of a qualified product shall require requalification. A document specifying the composition shall be forwarded to the US Army Public Health Center, Occupational Health Sciences Group, ATTN: MCHB-TS-O, 8252 Blackhawk Road Aberdeen Proving Ground, MD 21010-5403

3.2.2 Prohibited materials. The cleaning compound shall not contain any hazardous compounds as defined in 40 CFR 261, toxic pollutants in 40 CFR 401.15, nor hazardous air pollutants in 40 CFR63. The cleaning compound shall not contain detectable amounts of any of the following: abrasives, chromates, cadmium, lead, mercury, phenols, cresols, chlorinated solvents (except SNAP approved), except where specified within this specification. The following materials are prohibited unless they are being used as an essential active ingredient in the cleaner: sodium chloride, urea, sodium sulfate, nitrites, nitrates, sucrose or any sugars.

3.2.2.1 Toxicity. The cleaning compound shall not contain any chemical listed by the current report of known carcinogens of the National Toxicology Program (NTP) (see 6.4.1). Any carcinogenic components in the cleaner in a concentration of 0.1 percent or greater by weight or volume will be regarded as sufficient for the presence of a carcinogen in the cleaner. The cleaners supplied under this specification shall have no adverse effect on human health (see 6.4.1 and 6.5) when used for their intended purpose (see 6.1). Army toxicity questions and/or a toxicity clearance request should be addressed to: Commander, US Army Public Health Center.

3.2.2.2 Hazardous Air Pollutants. The cleaner shall contain no chemicals listed as Hazardous Air Pollutants (Section 112 (B) HAP List) (see 6.4.2). Any HAP components in the cleaner in a concentration of 1.0 percent (%) or greater by weight (wt.) or volume (vol.) will be regarded as the presence of a HAP in the cleaner. The product containing less than 1 % of HAP shall be considered as a HAP free cleaner.

3.2.2.3 Documentation. A certified letter from the US Army Public Health Center, confirming the requirements of this section (3.2.2) shall be sent to the DEVCOM Army Research Laboratory, Weapons and Materials Research Directorate, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

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3.3 Performance and physical requirements. The cleaning compound shall be when tested in accordance with and meet all the requirements listed in table I.

3.4 Cleaning method. The cleaning method shall be identified as hand wipe. Appropriate materials used for wipers shall be described by the manufacturer on the Technical Data Sheet for the product.

3.5 Storage/shelf life. Acceptable storage shelf life shall be a minimum of 1 year.

3.6 Disposal. Where possible, these cleaners should be recycled or biodegraded. Disposal of used cleaner should be in accordance with local, state and Federal regulations. Care should be taken to avoid mixing used cleaner with other waste materials, especially those containing halogenated solvents.

3.7 Standard contaminant. The standard contaminant used within section 4 shall be prepared by mixing two parts by weight of hydraulic fluid (MIL-PRF-83282 or equivalent), one part by weight lubricating grease (MIL-PRF-81322 or equivalent), and one-tenth part by weight of carbon black. This mixture shall be applied to panels, baked for two hours at 55°C (130°F), and cooled to room temperature. All cleaners must be qualified to the standard contaminant. If other contaminants are agreed upon the standard contaminant must also be inspected and reported as the baseline for performance data across all qualified cleaners.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)
b. Retention inspection (see 4.3)

4.2 Qualification inspection. The qualification inspection shall consist of all tests specified in table I. Failure of any test shall be cause for rejection.

4.2.1 Qualification samples. The initial qualification samples shall consist of the required amount of the cleaning compound that is needed by the testing facilities. The cleaning compound shall be of the class, and type to be used in filling contract orders. Samples shall be identified (see 6.3.1.1) and forwarded to the appropriate testing facility, and the subsequent testing reports (see 4.2.2) shall be forwarded to the qualifying activity (see 3.1.2) for retention.

4.2.2 Test reports. The manufacturer shall provide certified test reports from the agreed upon ISO 9000/IEC 17025 or NADCAP certified laboratory showing methods used and testing results. The material must conform to all the requirements of this specification. The reports shall be submitted to the DEVCOM Army Research Laboratory, Weapons and Materials Research Directorate, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

4.3 Retention inspection. To ensure compliance with all the requirements of this specification the

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manufacturer shall complete DD Form 1718 (Certification of Qualified Products) for each plant location and certify that the statements listed on the DD Form 1718 are true. Unless otherwise specified by the qualifying activity (see 6.2), the time of periodic verification by certification shall be in two-year intervals from the date of original qualification (see 3.1.1).

4.4 Testing methods. The tests of this specification shall be conducted in accordance with the applicable methods specified in table I and listed in 4.4.1 to 4.4.6. The contents of each selected container for sampling shall be thoroughly mixed in accordance with manufacture's recommendations.

TABLE I. Properties and test methods.

TEST TYPE	TEST METHOD and/or REFERENCE PARAGRAPH	REQUIREMENT
Effect of Cleaners on Unpainted Surfaces	ASTM F485 in accordance with Paragraph 4.4.1	Water-break free in accordance with Paragraph 4.4.3
Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces	ASTM F502 in accordance with Paragraph 4.4.2	No visual change in color or softening of coating
Shelf-life	See 6.2 and 6.6	≥1 year
Water-break free	ASTM F22 in accordance with Paragraph 4.4.3	4.4.3
Sealant Adhesion	See Paragraph 4.4.4	4.4.4.c
Adhesive Bonding	See Paragraph 4.4.5	4.4.5.h
Paint Adhesion	See Paragraph 4.4.6	ASTM D3359: Rating ≥4B

4.4.1 Effects on unpainted surface. The alloys listed in table II shall be verified per ASTM F485. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration.

TABLE II. Metal coupon material requirements.[1/]

Material	Material Form[2/]	Specification and Temper
Al 2024	Sheet / Strip	AMS QQ-A-250/4 and Temper T3
Steel 1010	Sheet	---
Steel 4340	Sheet / Strip	AMS 6359 / AMS-H-6875

[1/] If the material is specified for a test, then this table defines the requirements for the material.

[2/] Cut oversize coupons from bar or rod and grind to finish dimensions if necessary.

4.4.2 Effects on painted surfaces. The cleaner's effect on painted surfaces shall be verified per ASTM F502 for all coating systems listed in table III. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration. Any visual change in color or softening of coating constitutes a failure. Test panels shall be made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and treated per the requirements found in table III.

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TABLE III. Coatings systems for test by effects on painted surfaces ASTM F502 & adhesion tape test per ASTM D3359.

COATINGS TO BE TESTED		SURFACE TREATMENT
PRIMER	TOPCOAT	
NONE	NONE	MIL-DTL-81706 Type II (TCP)
MIL-PRF-23377 Class N	NONE	MIL-DTL-81706 Type II (TCP)
MIL-PRF-23377 Class N	MIL-DTL-53039 or MIL-DTL-64159	MIL-DTL-81706 Type II (TCP)

4.4.3 Water break free. The procedure described below (a – c) is carried out in order to prepare panels to be tested by the water break procedure, ASTM F22.

a. Test panels shall be 1 x 70 x 150 mm (0.04 x 2.75 x 6 inches) made from 2024-T3 per SAE AMS QQ-A-250/4 sheet.

b. Apply the standard contaminant mixture (paragraph 3.7), or other contaminants appropriate to the specific application, to three (3) test panels. Wipe the panel surfaces with a cleaning cloth to remove gross contaminants.

c. Wipe each test panel five times with a clean cloth soaked with the candidate cleaner.

4.4.4 Sealant adhesion. Sealant adhesion shall be verified by following the peel test procedure described below (a - c). Test panels shall be 1 x 70 x 150 mm (0.04 x 2.75 x 6 inches) made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and conversion coated per MIL-DTL-81706, type I, class 1A. There shall be a minimum of eight test panels tested per primer.

a. Apply each of the primers listed in table IV to the properly prepared test panels per the primer specification. Scuff sand the primed panels with abrasive mats conforming to A-A-58054, type I, class 1, grade A or equivalent. For each primer, follow the procedure described below.

b. Apply the standard contaminant mixture (paragraph 3.7), or other contaminants appropriate to the specific application, to eight (8) test panels. Wipe the panel surfaces with a cleaning cloth to remove gross contaminants.

c. Examine/compare and record failure surface characteristics, particularly adhesive/ cohesive failure modes. Record measured peel strength values and any pertinent observations or anomalies.

TABLE IV. Primers for Sealant Adhesion and Paint Adhesion.

Specification	Description
MIL-PRF-23377 Class N	Epoxy, High-Solids: NON-CHROMATED
MIL-DTL-53022 Type IV	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free

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4.4.5 Adhesive bonding. Adhesive bonding shall be verified by following the procedure described below (a - h) using aluminum (7075-T6 bare), substrate. The bonding surfaces of the test panels shall be roughened by mechanical abrading.

a. Cut a sufficient number of test panels of each substrate to provide a minimum of six specimens per cleaner for each of the two testing temperatures (i.e. minimum twelve test specimens per cleaner per substrate). Use panel thickness and dimensions that are appropriate to fabricate test specimens per ASTM D3167. The flexible adherends shall be 0.63 mm (0.025 in.) and the rigid adherends shall be 1.63 mm (0.064 in.).

b. Prepare and apply the standard contaminant according to paragraph 3.7.

c. Remove the gross contaminant with a clean cloth.

d. Hand wipe the candidate-cleaner panel surfaces until visually clean with a clean cloth soaked with the candidate cleaner at the specified use concentration. For aqueous cleaners, hand wipe panels again with de-ionized water (ASTM D1193, Type IV) until visual evidence of the cleaner is removed. Hand wipe the control-cleaner panel surfaces until visually clean with a clean cloth soaked with methyl ethyl ketone (ASTM D740).

e. Abrade the panel bonding surfaces with an orbital sander: 180 grit for aluminum. Clean again per step d.

f. Fabricate test specimens per ASTM D3167, using a paste adhesive (Dexter Hysol EA 9309.3NA or equivalent). Use shims or scrim cloth, if necessary, to control the bondline thickness to 5 - 10 mils. Cure the adhesive as recommended by its manufacturer. Record the bondline thickness and the adhesive cure conditions.

g. Perform floating roller peel tests in accordance with ASTM D3167 on a suitable testing machine (e.g., Instron or equivalent). Condition and test at least six (6) specimens for each testing temperature as follows: (1) Condition at 25° C (77° F) / ambient humidity for 24 hours minimum. Test at 25°C (77° F). (2) Condition at 80° C (176° F) / 95% humidity environment for 30 days. Test at 80° C (176° F).

h. Examine/compare and record failure surface characteristics, particularly adhesive/ cohesive failure modes. Record measured peel strength values and any pertinent observations or anomalies. Record the contaminants used, if other than the standard contaminant, and the cleaner concentration used (for aqueous cleaner).

4.4.6 Paint adhesion. Paint adhesion shall be verified by following the procedure described below (a - e):

a. Prepare 3 x 100 x 100 mm (0.125 x 4 x 4 inch) aluminum test panels made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and conversion coated per MIL-DTL-81706, type I, class 1A. There shall be a minimum of three test panels tested per primer. Allow panels to dry for 24 hours at room temperature.

b. Apply the standard contaminant mixture (paragraph 3.7), or other contaminants appropriate

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to the specific application, to the test panels.

c. Clean the panels with the candidate cleaner using the cleaning process that is recommended for the particular cleaner/application. Allow panels to dry at room temperature for 24 hours.

d. Apply each of the primers listed in table IV to the properly prepared (soiled and cleaned) test panels per the primer specification. The primer coatings shall be allowed to air dry for not less than 14 days, or air dry for not less than one hour followed by 24 hours at 65.5 ± 3°C (150 ± 5° F) prior to testing. Then immerse the test panels in de-ionized water for 24 hours.

e. The paint system shall be tested after drying in accordance with applicable paint specifications. Each sample shall be tested using ASTM D3359, method B using any commercially available tape (1 inch width) that will yield a minimum of 80 oz of adhesive resistance (and can be verified with OQE) over the tested coating when tested in accordance with ASTM D3330/D3330M. CARC adhesion shall not be less than scale 4B with any layer of the system. Nonconformance shall constitute failure of this test. For a current listing of approved adhesion test tapes: Contact the DoD CARC Commodity Item Manager at CCDC Army Research Laboratory, FCDD-RLW-MC, ATTN: Camouflage, Coatings and Corrosion Team, 6300 Rodman Road, Building 4600, Aberdeen Proving Ground MD 21005.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cleaning compounds covered by this specification are intended to be used for hand wipe cleaning on metallic substrates prior to painting, metal surface treatment, non-structural sealant application and adhesive bonding. This specification applies to end items/components and equipment as defined by National Emission Standards for Hazardous Air Pollutants (NESHAPs) for DoD assets. Cleaners approved to this specification are not intended for cleaning prior to die-penetrant inspection (FPI) unless specifically approved for FPI by the responsible technical authority. These cleaners should only be used when documented in approved DOD technical publications for maintenance of DOD materiel.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Specify class, type, grade, and category of the cleaner (see 1.2).
c. If issues of documents are different (see 2.2.1, 2.3, and 2.4).
d. If the time of periodic verification is different (see 3.1.1 and 4.3).
e. Specify shelf life requirements (see table I and 6.6).
f. If packaging requirements are different (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products Database QPD-32405, whether or not such products have actually been so listed by that date. The attention of manufacturers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information (questions, etc.) pertaining to qualification and testing of products and the certified test results from the approved testing facility for the qualified products database (QPD) testing (see 4.2) and should be submitted to the Coatings and Corrosion Team, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

6.3.1 Inspection and other information. In addition to the qualification test samples, the qualifying activity will request the manufacturer to submit to the qualification activity: (a) a certified test report showing that the material conforms to the requirements of this specification (see 4.2.2); (b) one copy of the SDS (see 6.5); (c) certification that the manufacturer's material meets the HAP and VOC content requirement (see table I, 1.2.2 and 3.2.2).

6.3.1.1 Qualification inspection sample identification. Qualification inspection samples are to be forwarded to the laboratory designated in the letter of authorization (see 6.3) and identified as follows:

QUALIFICATION TEST SAMPLES

Specification MIL-PRF-32405, Class 1, 2, or 3, Type I, II, or III (as applicable)

CLEANER, HAND WIPE, METALLIC SUBSTRATES, LOW OR EXEMPT
VOC AND HAP FREE

Manufacturer's name and product number
Submitted by (name and date) for qualification testing in accordance with authorization
(reference authorizing letter)

6.4 Definitions.

6.4.1 Carcinogens. The Occupational Safety and Health Administration (OSHA) definition of carcinogens is those chemicals / processes appearing in lists 1, 2A, and 2B of the International Agency for Research on Cancer (IARC) [1/]; substances known to be carcinogenic and occupational

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exposures associated with a technological process known to be carcinogenic by the National Toxicology Program (NTP) Report on Carcinogens (latest annual report) [2/]; and OSHA regulated carcinogens [3/].

[1/] Copies of these volumes may be found in medical libraries or through the World Health Organization, 1211 Geneva 27, Switzerland.

[2/] Copies may be obtained from the Public Health Service, National Toxicology Program, Public Information Office, P.O. Box 12233, MD B2-04, Research Triangle Park, NC 27709 or at http://ntp-server.niehs.nih.gov.

[3/] Copies may be found in 29 CFR part 1910 Subpart Z.

6.4.2 Hazardous Air Pollutant (HAP). HAP is defined as any substance listed under Section 112 of the Clean Air Act or its modifications. The text of the Clean Air Act, listed pollutants and modifications are kept by the Environmental Protection Agency (EPA) and are accessible through the website: http://www.epa.gov.

6.4.3 Volatile organic compound (VOC). VOC is defined in 40 CFR Part 51.100. Volatile organic compounds (VOC) means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions. For the latest complete definition of VOC see: http://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm.

6.4.3.1 Exempt VOC. A list of exempt VOC is listed in 40 CFR Part 51.100 (s). For the current exempt organic compounds see: http://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm. Compounds that are explicitly exempted from regulation by the EPA as VOCs are those compounds of carbon that have negligible photochemical reactivity (they do not produce ozone in atmospheric reactions).

6.5 Safety Data Sheet. A SDS must be prepared and submitted in accordance with FED-STD-313. The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Questions pertinent to the effect(s) of these cleaners on the health of personnel using them should be referred by the procuring activity to the appropriate medical service, who will act as its adviser. Contracting officers will identify the activities requiring copies of the SDS.

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27- M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order:

(1) The Inventory Control Points that manage the item and
(2) The DoD Service and Agency administrators for the DoD Shelf-Life Program.

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Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.logtool.com/Toolbox/dod-shelf-life

6.7 Subject term (key word) listing.

Aqueous Environmental	Non-hazardous
Non-aqueous	Support vehicles

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY – MR

(Project 6850-2021-009)

Review activities:
Army - AT, AV, EA, GL, MI
DLA - GS, GS3
GSA - FAS
MISC - MP

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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